

Sebastian Horta

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EDUCATION

Columbia University (SEAS)

New York, NY

B.S. Computer Science, Minor: Electrical Engineering

Aug 2023 – May 2026

- Coursework: Computer Networks, NLP, Adv. C++ Systems Programming, Competitive Programming, Artificial Intelligence, Data Structures & Algorithms, Systems Programming, CS Theory, UI Design, Computer Systems
- Resident Adviser: Organized 10+ community events for 700+ residents, leveraging cross-functional teamwork

EXPERIENCE

Machine Learning Engineer — Columbia University Robotics Lab

2025

New York, NY

- Built **ML** pipeline in **Python** to detect gait events (heel strikes/toe offs) from VR motion-tracker data
- Engineered unsupervised **KMeans** clustering to classify VR trackers by body segment and foot side
- Developed **cross-correlation-based synchronization** between VR tracker streams and MAT ground truth
- Trained **Random Forest** classifier, correctly detecting ~85% of gait events from VR tracker data

Full Stack Software Developer — Freelance

2025 – 2026

Remote

Web Archiver Tool | React + FastAPI

2025

- Built offline web archiver with structured storage and versioning for **research** and offline site analysis
- Developed **recursive crawler** using **requests** + **BeautifulSoup** for HTML and asset capture
- Implemented URL rewriting and error handling for irregular link structures
- Created **React** dashboard with real-time progress tracking and iframe previews

Client Business Website (Live Site) | Next.js + TypeScript + Tailwind

2026

PROJECTS

Allowance | Python

2026

- Designed a **P2P blockchain** enforcing AI agent spending limits via **ECDSA**-signed transactions and **PoW**
- Implemented **Merkle tree** transaction integrity and fork resolution to maintain consensus across 4+ nodes
- Built a peer discovery tracker with heartbeat monitoring, persistent **TCP** connections, and thread-safe state
- Validated correctness with **106 tests** covering transaction signing, overspend rejection, and tamper resistance

C++ Generic Database & Build Tool | C++

2026

- Built a reusable **C++ database library** using class templates to store and retrieve any data type from disk
- Implemented a **build automation tool** supporting Makefile parsing, variable substitution, and implicit rules
- Resolved build dependencies recursively with **cycle detection**, only rebuilding targets whose source files changed
- Accelerated startup by caching parsed build rules to disk, skipping re-parsing when the Makefile is unchanged

SKILLS

Languages & Frameworks: Python, Java, C/C++, JavaScript/TypeScript, HTML/CSS, React, Next.js, Tailwind

Backend & AI/ML: FastAPI, Flask, Node.js, REST APIs, SQL, scikit-learn, pytest, OpenCV, LLMs

Tools: Git, Unix/Linux, CI/CD, Figma

LEADERSHIP & ACTIVITIES

SHPE at Columbia – Committee Board Member – Planned cultural and career events, resulting in increased presence on campus

IEEE, ColorStack, CU ADI, CU BJJ Club, Sabor Dance Club – Member – Engaged in club events, community building, and collaborative projects

Collegiate Science and Technology Entry Program (CSTEP) – Member – Participated in program initiatives and STEM outreach activities

AWARDS

Columbia Spirit Award (2026) • Anders Scholarship (2024) • Columbia University Nathaniel Arbiter Scholarship (2024)

FSC President's List (2022–23) • NYS Scholarship for Academic Excellence (2022)